

Similarity Score

CITS3001 Advanced Algorithms — Time limit: 3s

The unit coordinator suspects two submissions are a little too similar. As a first, crude measure, compute the length of their *longest common subsequence*: the longest string that can be obtained from *both* submissions by deleting zero or more characters (without reordering the rest).

Input

The input consists of two lines, each containing a non-empty string of at most 1000 lowercase letters (a–z).

Output

Output a single integer: the length of the longest common subsequence of the two strings.

Sample Input 1

```
dynamic
programming
```

Sample Output 1

```
3
```

Sample Input 2

```
abc
xyz
```

Sample Output 2

```
0
```